

Evaluating and validating Claude output — Principles and critical tips

Pair: [evaluating-validating-output-references.md](#)

Course: Associate · Module 3 · Evaluating and Validating Claude's Output

Related: Module 2 prompting stack under [claude/prepcourses/foundations/module2/](#)

Goals this note serves: AI as operator · bug bounty / appsec · audit work · architect judgment

Scope: authorized study and authorized testing only

One-sentence definition

Discernment checks AI output against requirements, sources, and professional standards.

Diligence sets when human review is mandatory and keeps accountability on you when you ship.

Why it exists

Time saved is small and visible. Cost of an unverified error is large and arrives later. One fabricated figure in a client deck can cost more trust than the minutes the draft saved. You own every claim you ship, whether you wrote the words or the model did. This module is the largest exam section because **accountability** is the product.

Cautionary pattern

A consultant asked for market-sizing stats. Four figures were sound. One growth rate was fabricated and still looked clean. The client analyst caught it in the room. Ten minutes of "research" became a week of trust repair. Nothing on the screen looked wrong. That is the failure mode.

Mechanism (first principles)

1. Fluent text is not evidence. The model uses the same confident voice for a fact and a guess.
2. Evaluation is a **protocol**, not a feeling. Run the same three references every time.
3. **Accuracy** asks if present claims are correct. **Completeness** asks if anything critical is missing. They fail separately.
4. Prevention in the **prompt** (uncertainty allowed, source limits, auditable cites) beats cleanup after ship.
5. **Stakes** set review depth. Casual review on zero-tolerance work is the main process failure.
6. Format is a **reliability** choice. Prose numbers are guesses in answer shape. Code execution returns a checkable computation.

High-level map

AI draft

- set stakes (depth of review)
- check requirements + sources + professional standards
- scan failure signatures (hallucination, inconsistency, bias, completeness)

- triage: ready | revise | human override
- if high stakes: ground prompt, recompute figures, escalate by policy
- edit for audience + pick format by reliability
- ship only if you can stand behind every claim

Part	Job
Discernment	How you review (three references + completeness)
Failure patterns	What to spot fast (signatures, not "looks wrong")
Fact-check and grounding	How to prevent and audit
Diligence	When human review is non-negotiable
Editing	Clarity, tone, format for the reader
Output formats	Inline, artifact, structured, code-executed

Fundamental unit: one **claim** that will leave the chat.

Mechanism: verify the claim against source or recompute it, or do not ship it.

Discernment: three evaluation references

Reference	Question	Failure example
Requirements	Does output match what you asked, not only the easy parts?	Missed one competitor in a three-competitor ask
Source material	Do claims match uploaded or supplied docs?	"\$40/user" without "minimum 10 seats" from the PDF
Professional standards	Does this pass in your field?	Number without units, recommendation without reasoning, untraceable cite

Stakes calibration

Stakes	Review depth
Zero-tolerance (legal, financial figures, compliance)	Verify every claim. Accuracy beats speed.
Low-stakes internal brainstorm	Lighter review is fine.
Risk	Same casual review on both domains

Set stakes **before** you set review depth.

Three-way triage

Verdict	When
Ready to use	Meets requirements, matches sources, clears standards. Ship.
Needs revision	Close. One specific gap remains. Note the gap. Iterate.
Needs human override	Stakes, errors, or uncertainty block ship on draft alone. Escalate.

Document the reasoning. Do not only feel "looks good."

Completeness is a separate pass

Accurate text can still omit the one factor that changes the decision. Missing items are harder to see than wrong items. Nothing on the screen points at silence.

Three protocol examples

Output	References result	Verdict
Competitor pricing from uploaded PDFs	Clean form, but one price drops seat minimum from source	Needs revision (source-restricted re-prompt)
Three options to cut invoice time (internal)	Requirements met, no source file, low stakes	Ready to use (do not over-verify)
Compliance gap vs regulation not uploaded	Confident gaps from training recall, regulatory stakes	Needs human override (expert must own)

Polish is not the signal. References and stakes are the signal.

Hallucinations, inconsistencies, and bias

Plausible is not verified.

Hallucination signatures

Pattern	Signature
Plausible-but-unsupported	Fits the topic. No basis in source or fact. Looks fine.
Fabricated specifics	Invented stats, dates, names, quotes, citations. Precision fakes authority.
Confident tone masks uncertainty	Guess and grounded fact share the same assured voice.

Tell for uncited precision: real figures this specific usually ship with a source. An uncited "63 percent" is often a number-shaped guess.

Inconsistency and bias

Pattern	What to do
Internal contradiction (long docs)	Hold whole document in view. Run a consistency pass, not only paragraph skim.
Confirmation bias in framing	If the prompt leans one way, watch for over-agreement on open questions.

Completeness failure pattern

Compare many files. Output lists differences and feels thorough. It missed differences in the **most important** file. Completeness fails where attention is lowest. A confident summary of easy files masks silence on the hard one.

Capability hallucination

Claims like "I emailed your team" or "I saved the file" without a tool that can do that are unverified. In the product, Claude works with conversation, connected tools, and uploads. It does not perform external actions without a granted tool. Confirm external actions in the real system.

Fact-checking and grounding

Strongest verification lives in the prompt, before the draft exists.

Prompt for verifiability

Habit	Effect
Permit "I do not know"	Cuts invented fill under pressure to answer
Restrict to provided sources	Bounded retrieval. Flag what sources do not cover.
Require auditable citations	Source + location you can open. Untraceable cite is not a cite.

Grounding techniques

Technique	Use
Quote first, then analyze	Extract supporting quotes before conclusions
Best-of-N comparison	Re-run. Agreement raises confidence. Divergence marks soft spots.
Validate against authority	High-stakes claims need a trusted external reference, not only a second model pass
In-product aids	e.g. cell-level citations in spreadsheet tools when available

Paste-ready prompt fragments (course skill)

Permission to not know

If the answer is not supported by the documents I provided, say so explicitly rather than estimating. It is acceptable to answer "the provided materials do not cover this."

Source restriction

Answer using only the attached contract. Do not use general knowledge. For anything the contract does not address, list it under "Not covered by this document."

Auditable citation

For every claim, cite the section and clause number it comes from, in parentheses, so I can verify it against the source.

Quote-grounding

Before you analyze, extract the exact sentences from the document that bear on my question. Then base your analysis only on those quotes.

Verification checklist (before you rely on output)

1. Did I allow uncertainty?
2. Did I restrict to sources when appropriate?
3. Did I require citations I can audit?
4. Did I check high-stakes claims against something authoritative?

Diligence: when human review is non-negotiable

Some outputs must never leave as a solo model draft, no matter how good they look. Set thresholds **in advance**. Policy beats panic.

Four risk thresholds

Threshold	Question
Stakes	Cost if wrong?
Reversibility	Can you undo? Sent client pack vs private draft
Audience	External, executive, regulatory vs internal working
Regulatory exposure	Rule, contract, or law in play? AI does not remove obligations

Do-not-ship-without-review list (decide ahead)

- Final client deliverables
- Audit-critical or financially material calculations
- Regulated, confidential, or highly sensitive data
- Public or legal communications with lasting consequence

Iteration versus escalation

Productive iteration improves each round. When rounds stop improving, stop prompting. Bring a human expert. More prompts do not manufacture required judgment.

You own the output

Accountability does not transfer to the tool. Shipped work with Claude is your work. Standards match unaided work.

Escalation scenarios

Case	Thresholds	Action
Fast "yes" — internal agenda	Low stakes, reversible, internal	Ship. No escalation.
Deceptive "looks fine" — board financial summary	High stakes, executive, partly irreversible	Human review. Recompute figures with code.
Slow creep — five proposal rounds, flat improvement	External high stakes + diminishing returns	Stop. Colleague fresh read.

Editing and adapting for audience

Claude drafts. You deliver. Accurate is not finished.

Three edit passes

Pass	Job
Clarity	Cut hedging. Tighten. Remove dead weight. Prefer precision over thorough fluff.
Tone	Match peer, client, or regulator register.
Formatting	Scannable for exec. Detail for team. Clean for external.

Audience calibration

Same facts, different selection and depth:

Audience	Shape
Executive	Decision and impact first
Working team	Method, detail, next action and owner
External	Disclosure control and framing

Compare before you lock a base

When quality matters, generate more than one draft (runs or models). Pick the strongest base to edit. Comparison is cheaper than rescuing a weak first draft.

Same 18 percent, two audiences (pattern)

- **Exec:** "Q3 processing time rose 18 percent, driven by volume plus onboarding three new hires. Recommend standardized onboarding docs."
- **Team:** Keep drivers, add action, owner, and timeline.

Neither is the raw long hedge paragraph.

Choosing output formats

Format is a **reliability** decision.

Format	Use
Inline	Conversational answer you act on inside chat
Artifacts	Standalone document or code you refine and reuse
Structured	Tables and schemas for tools or readers
Code execution	Numbers that must be right — compute, do not invent

Prose vs code-executed numbers

Path	Result
Prose total	Fluent. Fast. Often unverifiable. Wrong total poisons every slide.

Path	Result
Code on the file	Computed sum, ranked accounts, chart. Traceable to rows.

Determinism attaches to the **executed** calculation. The model writes the code, so the logic can still contain a bug. You can read, verify, and re-run. That is the trust path for reported figures.

Low-stakes gut-check can use prose. Reportable decisions need code execution when numbers matter.

Curate inputs

Technique	Why
De-duplicate sources	Avoid three near-copies that force false reconcile
Label each input role	"Approved policy" vs "draft responses"
Prune noise	Noise in → noise out

Clean, labeled, minimal inputs beat a large undifferentiated pile.

Operator triage card (exercise pattern)

For each AI output before ship:

1. Stakes? (low / medium / zero-tolerance)
 2. Requirements complete?
 3. Sources match?
 4. Professional standards?
 5. Completeness pass (what is missing)?
 6. Failure signatures (fabricated specific, contradiction, over-agreement, capability claim)?
 7. Verdict: ready / revise / human override
 8. If numbers material: code path used?
 9. Audience edit done?
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Critical points everyone misses

1. **Accountability stays with you.** Exam weight matches that fact.
2. **Completeness fails silently.** Scan for silence, not only wrong text.
3. **Uncited precision is a tell.**
4. **Long docs need a consistency pass.**
5. **Set do-not-ship list before the moment.**
6. **Diminishing returns mean escalate, not re-prompt.**
7. **Format by reliability, not by prettiness.**

Gotchas

If you see...	Then...
Fluent generic market stats	Demand source or mark unverified. Do not paste into client decks.

If you see...	Then...
Compliance analysis without uploaded regulation	Human override. Training recall can be stale.
"I emailed / saved / filed that"	Confirm in the real system or treat as false.
Board numbers from prose	Recompute with code. Human review.
Five flat iteration rounds	Stop. Fresh human read.
Thorough-looking multi-file compare	Re-check the hardest or most important file.

Critical tips for operator goals

Goal	Critical tip
Bug bounty / appsec	Require source-restricted notes on scope docs. Fabricated CVE or "fixed in version X" without advisory is a ship block. Completeness: did it miss the auth surface you care about?
Cloud / Azure-style audit	Material findings need evidence path (portal, CLI, log). Regulatory and client reports = human override list. Recompute counts and percentages with code when possible.
AI as operator	Pre-send grounding habits (I do not know, sources only, cites). Post-receive: three references + triage. Never skip stakes calibration.
Architect judgment	Trade-off recommendations need reasoning and alternatives. Confirmation bias: if you asked for one design, force a counter-option pass.

IF / THEN

IF	THEN
Deliverable is external or regulated	Human review before ship
Claim is a specific number or date	Source or recompute. Else cut or flag.
Source was not provided for a domain rule	Do not treat training recall as current law
Output is internal brainstorm only	Lighter review. Do not spend audit budget.
Rounds no longer improve	Escalate. Stop prompting.
Numbers feed a decision	Code execution path

Do this (minimal practice)

1. Take one AI draft you almost shipped this week.
2. Run three references + completeness. Write a one-line triage verdict.
3. Add the four grounding fragments to your next high-stakes prompt.
4. Write your personal do-not-ship-without-review list (five lines).
5. For one spreadsheet task, force code execution instead of prose totals.

Out of scope here

- Official doc tables → **references** pair
 - Module 2 component stack (prompt construction) → module2 principles
 - Full product UI tour → current Anthropic help (re-fetch)
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Module takeaways (operator card)

1. You own every shipped claim.
2. Evaluate against requirements, sources, standards. Calibrate depth to stakes.
3. Plausible is not verified. Learn failure signatures.
4. Build verification into the prompt.
5. Know review thresholds in advance.
6. Pick format by reliability. Compute numbers when they must be right.